UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

TOPIC 4 :Work Products in RE
TOPIC 5: Practices for Model-Based Requirement Documentation (Part 1)

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Contents

- Work Products in Requirements Engineering
- Characteristics of Work Products
  - Abstraction Levels
  - Level of Detail
  - Aspects to be Considered
- To model system visions using Goal decomposition (AND/OR)
- To model business process model using swimlane diagram

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Objectives

By the end of this unit, students will be able to:

- Analyze the characteristics and abstraction levels of work products used in Requirements Engineering to determine their purpose and interrelationships.
- Design a Goal Model that captures stakeholder objectives and links them to system-level requirements.
- Construct a Business Process Workflow Model (Swimlane Diagram) to illustrate the "As-Is" and "To-Be" processes based on a given case study.

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4.1 WORK PRODUCTS IN REQUIREMENTS ENGINEERING

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Definition: Work Product

A work product is a documented, intermediate, or final outcome produced during a work process.
Synonym: *Artifact*

Note:
The term *work product* is preferred over *artifact* because it emphasizes that it is the direct result of activities carried out in a process.

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Work Product

A work product is a documented, intermediate, or final outcome produced during a work process.
Synonym: *Artifact*

- Different types of work products are used in Requirements Engineering (RE).
- They can be distinguished by their purpose, form of representation, size, life span, and storage.

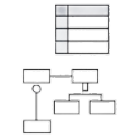
Frequently Used Work Products

Requirements	Work Products
Individual Requirements	individual requirements, user stories
A Set of Requirements	use cases, graphic models, task descriptions, epics
Requirements Documentations Structure	systems, business, stakeholder and user requirements specifications, product, sprint backlogs
Others	glossaries, textual notes

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Typical Work Products used in RE



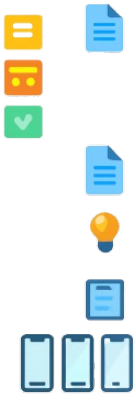
Work product	Purpose. The work product specifies /provides	Size
Single requirement		
Individual requirement	A single requirement, typically in textual form	S
User story	A function or behavior from a stakeholder's perspective	S
Set of requirements		
Use case	A system function from an actor's or user's perspective	S-M
Graphic model	Various aspects, for example, context, function, behavior (see Section 3.4)	M
Task description	A task that a system shall pertorm	S-M
External interface description	The information exchanged between a system and an actor in the system context	M
Epic	A high-level view of a stakeholder need	M
Feature	A distinguishing characteristic of a system	S-M

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Typical Work Products used in RE



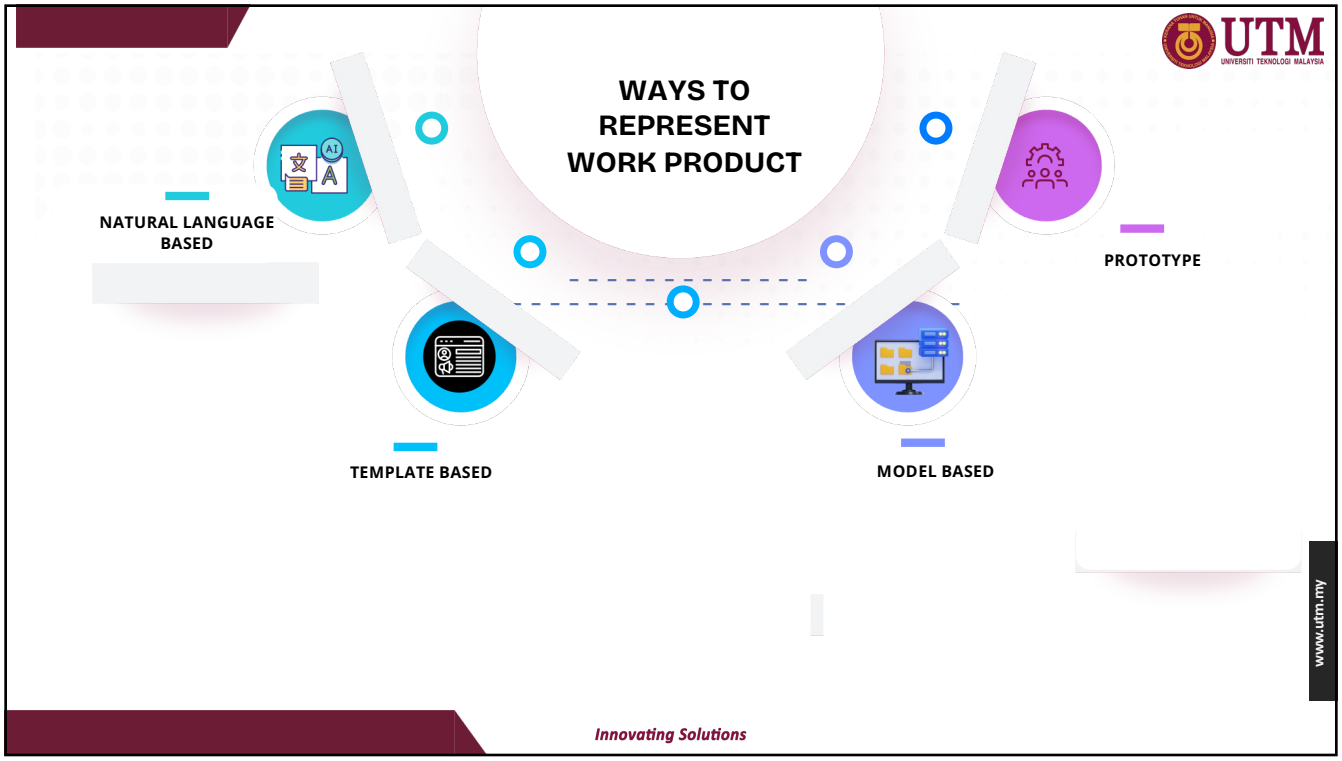
Work product	Purpose: The work product specifies/ /provides	Size
Documents or documentation structures		
System, business, stakeholder, or user requirements specica-tion	A comprehensive requirements document	L
Product and sprint backlog	A list of work items, including requirements	M-L
Story map	A visual arrangement of user stories	M
Vision	A conceptual imagination of a future system	M
Other work products		
Glossary	Unambiguous and agreed common terminolo-	M
Textual note or graphic sketch	A memo for communication and understand-ing	S-L

S: Small, M: Medium, L: Large, XL: Very large work product

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Characteristics of Work Products

- Most work products are **stored electronically** — as files, in databases, or within Requirements Engineering (RE) tools.
- Informal or temporary work products may also be kept in **other formats**, such as paper notes or post-it notes on a Kanban board.
- Every work product has a **lifespan**:
  - The time span from its creation until it is either discarded or becomes irrelevant.

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Categories of Work Products with respect to Life Span

1. Temporary Work Products
- Created mainly to facilitate communication and build shared understanding.

2. Evolving Work Products
- Developed and refined over multiple iterations.
- Require some metadata and may be subject to change control.

3. Durable Work Products
- Baseline or released versions.
- Require complete metadata and must follow a formal change management process.

Note:
- A temporary work product can be turned into an evolving work product by retaining it and adding metadata.
- Temporary or evolving work products can become **durable** once they are baselined or formally released.

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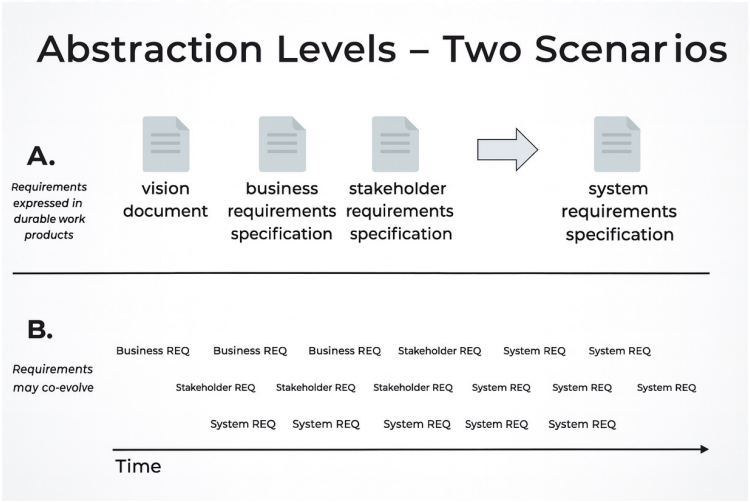
Abstraction Levels

- Requirements and their related work products exist at **different abstraction levels**.
  - They may range from **high-level requirements** describing a new business process
  - To **detailed requirements**, such as specifying how a particular software component should respond to an exceptional event.

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Abstraction Levels



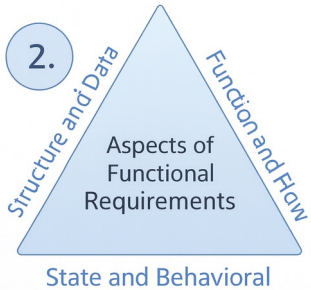
Scenario A: Sequential Refinement
- Requirements developed step-by-step in separate documents.
- Each document builds on the previous:
  - Vision Document → Business Requirements → Stakeholder Requirements → System Requirements
- Suitable for stable projects with clear goals (e.g., Waterfall model).
- Produces durable work products at each level.

Scenario B: Co-evolution of Requirements
- Requirements at different levels evolve together over time.
- Business, stakeholder, and system requirements are refined iteratively.
- Suitable for dynamic or uncertain projects (e.g., Agile approach).
- The timeline arrow at the bottom emphasizes continuous refinement — requirements are revisited and updated as understanding improves.

Aspects to be Considered in Work Products

1. **Kinds of Requirements**
   - Functional requirements
   - Quality requirements
   - Constraints

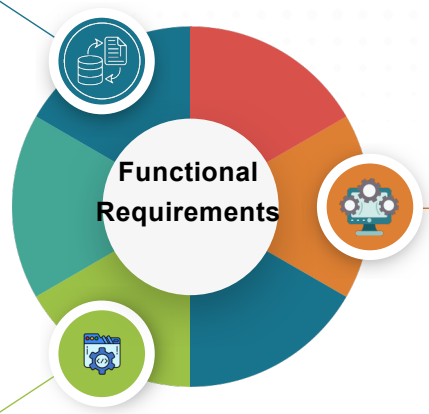
2.



3. **System Context and Boundary**
   - Requirements can only be understood in context
   - Domain requirements and domain assumptions in the system context

Cover all these aspects when documenting requirements and creating work products

Aspects of Functional Requirements



1

Structure and Data

- Static structure of a system and the (persistent) data
- What a system must know in order to perform the required functions and deliver the required results.
- Includes static relationships of data.

2

State and Behavioural

- State dependent behaviour of a system.
- How a system shall react to which external event

3

Functions and Flow

- Functions that a system shall provide.
- Flow of control and data within and between functions.
- What systems need for creating the required results from given inputs.

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5.1 MODEL-BASED WORK PRODUCT

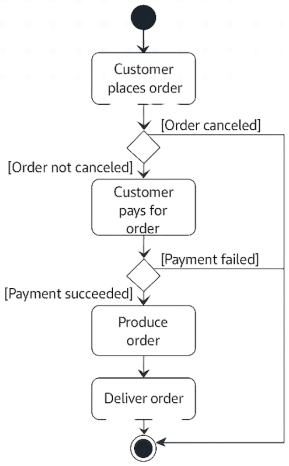
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Model-Based Work Product

Requirements represents in the natural language have limitations:
- Difficult to gain overview of a set of requirements and understanding the relationships between requirements

Textual requirements	Modelled requirements
REQ001: The customer places an order	
REQ002: The customer pays for the order	
REQ003: If the payment succeeded, than the order is produced	
REQ004: If the payment failed, then the order is canceled	
REQ005: After the order is produced it is delivered to the customer	
REQ006: The order can be canceled if the customer does not pay it	

Models in Requirements Engineering

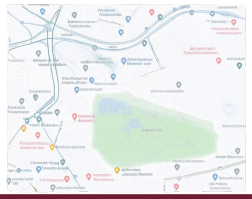
Definition

An abstract representation of an existing part of reality or a part of reality to be created

- Reality: City of KL



- Model: Map of KL



1. A model is made for a specific purpose
2. A model gives a representation of reality
3. A model is used to reduce information so that we can better understand reality or focus on part of the reality

3 Properties of Models

Role of Models in RE

- Models help to understand the relationships and interconnections between requirements.
- Models provide an overview of a set of requirements
  - by focusing on some aspects, for example behaviour.
  - and abstracting from all other aspects
  - supported by using a graphic notation for a model
- A graphical / diagrammatic representation of a model is called a diagram.
- Models may also be represented in a non-graphical way, for example with table.

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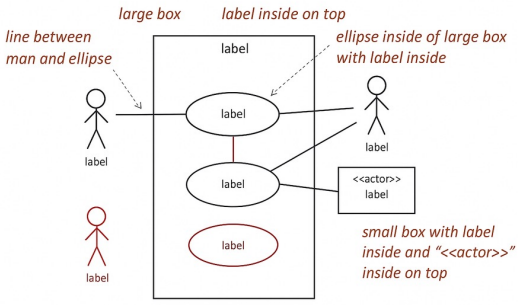
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Role of Models in RE

- Models and their diagrams are often based on a modeling language.
- A modeling language consists of grammatical rules (syntax) and a description of the meaning of the language constructs (semantics)
- Standardized modelling languages used for RE:
  - Unified Modeling Language (UML) - https://www.uml.org
  - Business Process Model and Notation (BPMN) - https://www.bpmn.org

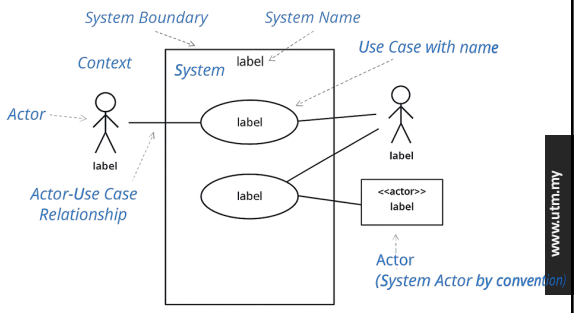
Syntax — Modeling elements/symbols and their valid combinations

(for Use Case Diagrams)



Semantics — Meaning of the modeling elements/symbols

(for Use Case Diagrams)



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USING MODELS ADVANTAGES & DISADVANTAGES



ADVANTAGES

- The elements and their connections are easier to understand and remember
- Focusing on a single aspect reduces the cognitive load for understanding the modeled requirements
- Requirements modeling languages have a restricted syntax which reduces possible ambiguities and omissions



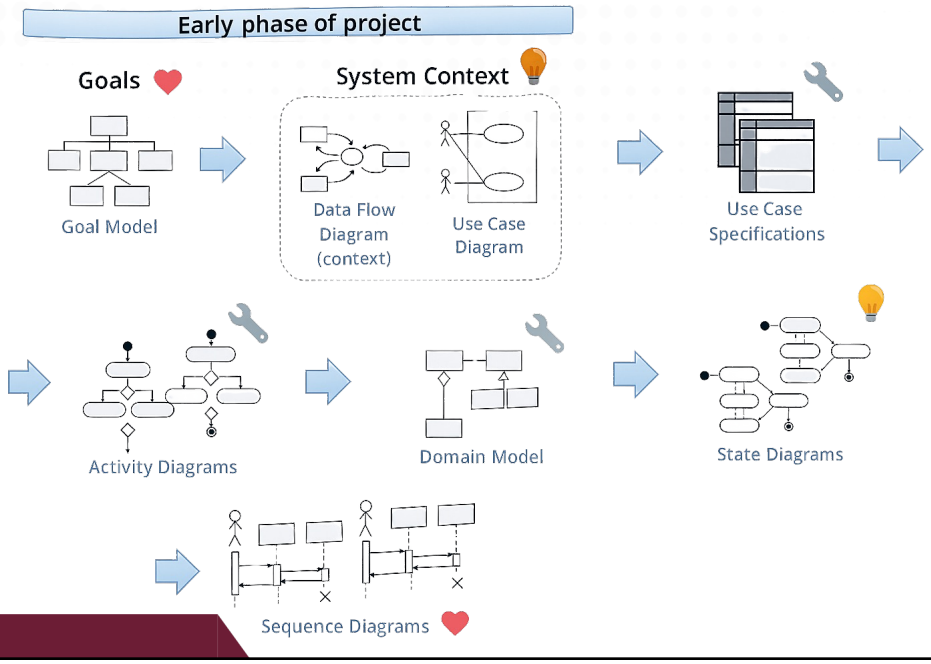
DISADVANTAGES

- Keeping models that focus on different aspects consistent with each other is challenging.
- Information from different models needs to be integrated for causal understanding.
- Models focus primarily on functional requirements.
- The restricted syntax of a graphic modeling language implies that not every relevant item of information can be expressed in a model.

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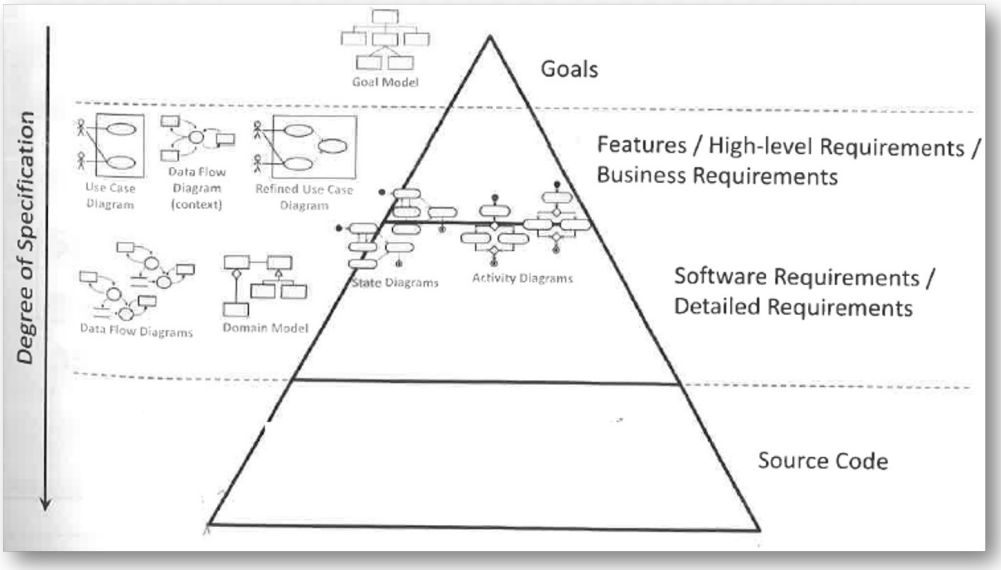
Model Overview



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Level of Specification



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5.1 (a) MODEL-BASED WORK PRODUCT

- GOAL MODEL

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Introduction to Goals

Definition of Goal

D A goal describes a **high level objective** of one or more stakeholders **about a property** of the **system to be developed** or the **development project**.

E Goals for a car navigation system:

- G_1: The system shall guide the driver to a desired destination automatically.
- G_2: The response time of the system shall be 20% lower compared with the predecessor system.

Goals for a university library system:

- G_2 : The system shall be used by students and employees.
- G_5: The system shall allow employees to use the library printers directly from their work place.

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Goal Modelling

Goal Model

D A **goal model** is a conceptual model that documents goals, their decomposition into sub-goals and goal dependencies.

- A goal describes an intention of a stakeholder
- Goals can be documented both in natural language and in the form of models
- An integral part of the documentation of goals is the description of refinement relationships (decomposition relationships) between higher and subordinate goals.

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Goal Model

- A hierarchy of goals
- Relates the high level goals to low level system requirements

- High level goals
  - Strategic concerns
  - For example: “Serve more passengers” for a train transportation systems
- Low level goals
  - Technical concerns
  - For example: “Acceleration command delivered on time” for a train transportation system

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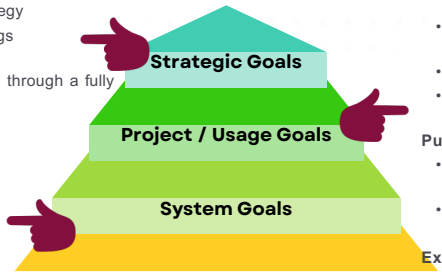
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Three Levels of Goals

- Describe the long-term vision and business objectives that drive the need for a system.
- They represent the why behind system development.

Purpose in Requirements Engineering:
- To align system development with organizational strategy
- To justify why the system exists and what value it brings

Example (UTM Library System):
- Enhance teaching, learning, and research excellence through a fully digital, user-centered library ecosystem



- Translate the strategic vision into specific operational outcomes or user-oriented objectives.
- They define what the organization or users expect to achieve through the system’s use.
- Owned by project teams, departments, or user communities
- Focus on outcomes of system use rather than internal functions

Purpose in Requirements Engineering:
- To define how success will be measured once the system is in use
- To capture stakeholder and user expectations in concrete terms

Example (UTM Library System):
- Provide seamless access to digital resources for students and researchers.

- Specify what the system must do or how it must perform to fulfill project or usage goals.
- They represent the technical realization of stakeholder needs.

Purpose in Requirements Engineering:
- To derive functional requirements (what the system shall do)
- To derive non-functional requirements (how well it shall do it)

Example (UTM Library System):
- Implement unified search, SSO login, and chatbot support

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Core Security Goals in SSDLC

- Embedding security practices throughout the Secure Software Development Life Cycle (SSDLC)
- Addressing potential vulnerabilities proactively and ensuring compliance with industry standards.
- Integration of CIA triad into each phase of the Software Development Life Cycle (SDLC)
- Crucial for building reliable and secure software systems.

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Core Security Goals in SSDLC

CIA TRIAD



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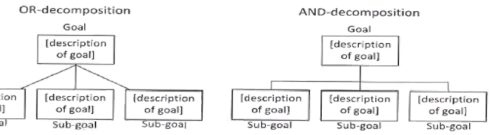
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Goal Models Representation

- Goal models describe the system through decomposition of the goals in sub-goals
- The goals models are represented as AND-OR trees
  - A goal is decomposed into sub-goals by AND / OR decomposition relationships

Graphical notation indicates type of decomposition (AND/OR).

- Notation for AND-Decomposition:
- Notation for OR- Decomposition:



- "AND-decomposition" (all sub-goals must be fulfilled in order to fulfill the higher goal (super-goal)
- "OR-decomposition" (at least one sub- goal must be fulfilled in order to fulfill the higher goal (super-goal)

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Goal Modelling

AND-Dependency (2)

AND-decomposition
OR-decomposition

G_1: comfortable and fast navigation to destination

$G_{1.1}$: automatic navigation
$G_{1.2}$: circumnavigating traffic jams
$G_{1.3}$: easy entry of destination

Goal Modelling

OR-Dependency (2)

AND-decomposition
OR-decomposition

$G_{1.2}$: circumnavigating traffic jams

$G_{1.2.1}$: manual entry of traffic jams in road traffic
$G_{1.2.2}$: autonomous update of traffic data

Goal Modelling

AND/OR Goal Trees

AND-decomposition
OR-decomposition

G_1: comfortable and fast navigation to destination

$G_{1.1}$: automatic navigation
$G_{1.2}$: circumnavigating traffic jams
$G_{1.3}$: easy entry of destination

$G_{1.2.1}$: manual entry of traffic jams in road traffic
$G_{1.2.2}$: autonomous update of traffic data

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5.1 (b) MODEL-BASED WORK PRODUCT

- **BUSINESS PROCESS WORKFLOW: SWIMLANE DIAGRAM**

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Business Process Workflow

What is a Business Process Workflow?

- A visual representation of how tasks, data, and decisions flow within or across organizational units.
- Defines "Who does What, When, and How."
- Helps ensure clarity, accountability, and consistency in business operations.

Importance in Requirements Engineering

- Provides a foundation for system modeling (e.g., use cases, data flow diagrams).
- Supports accurate requirements elicitation and stakeholder communication.
- Ensures new IT solutions integrate smoothly with legacy systems.
- Reduces project risk by clarifying business objectives early.

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Business Process Workflow

1

To understand current business processes before proposing IT solutions

2

To align system requirements with real business needs.



3

To identify roles, responsibilities, and interactions between departments or systems.

4

To recognize inefficiencies, redundancies, or bottlenecks in the existing proces

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Workflow Modeling

STEP 1

Frame the process

- Identify a set of related processes, including the target process to be improved, by developing an overall process map.
- Review all related organization documents and goals.
- Perform initial process assesment
- Identify individuals/roles that are a part of the process
- Important to make certain that you include everyone impacted

STEP 2

Understand the As-Is Process

- Map the current process workflow to show who does what, when
- The purpose is to identify:
  - Bottlenecks – where the process slows down
  - Role ambiguity – unclear responsibilities between staff or units
  - Duplicate efforts – redundant activities or approvals
  - Unnecessary steps – tasks that add no real value
  - Excessive handoffs – too many transfers between people or systems

STEP 3

Design To-Be (New) Process

- Decide on direction—abandon, stay as-is, improve, redesign?
- Develop ideas for characteristics or features of the new process
- Draw the to-be workflow.

STEP 4

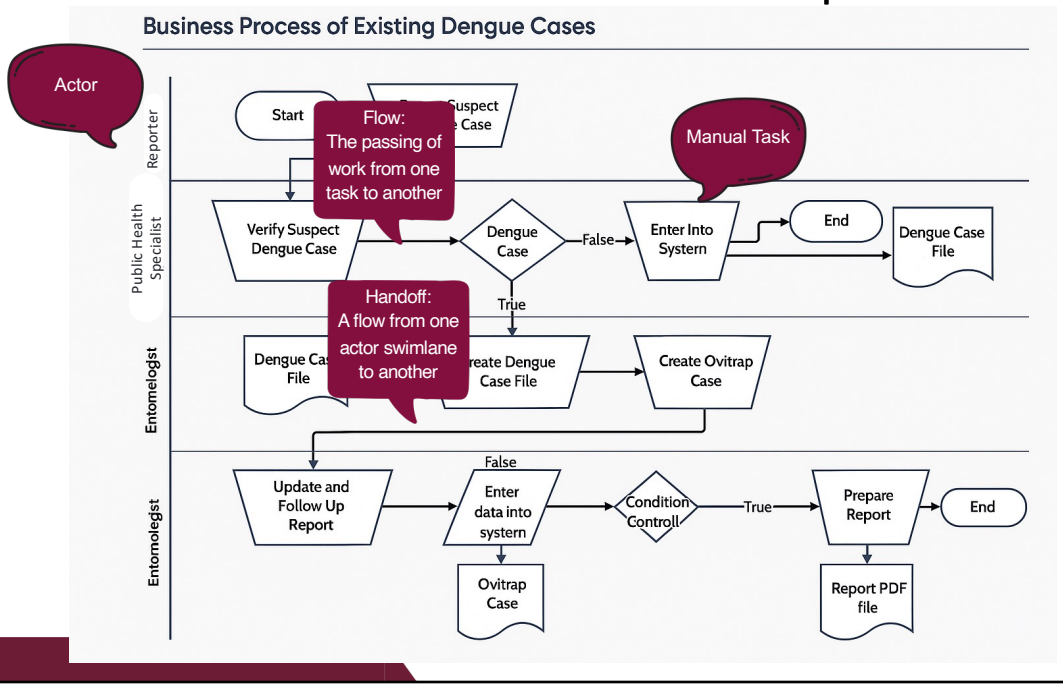
Develop Use Cases

- Model the process to be the use cases in the use case diagram

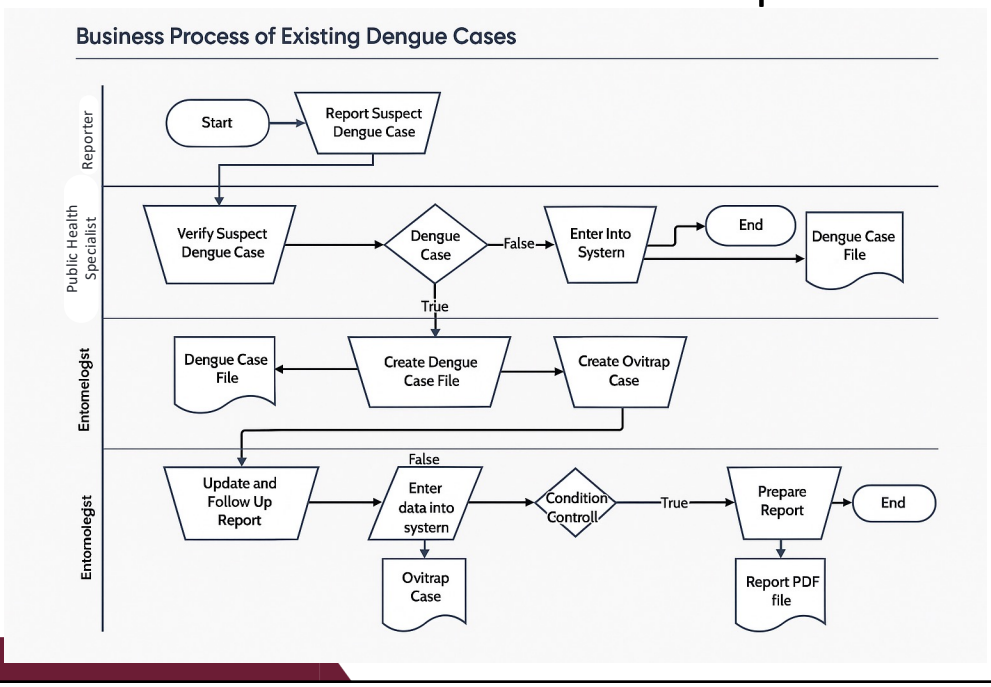
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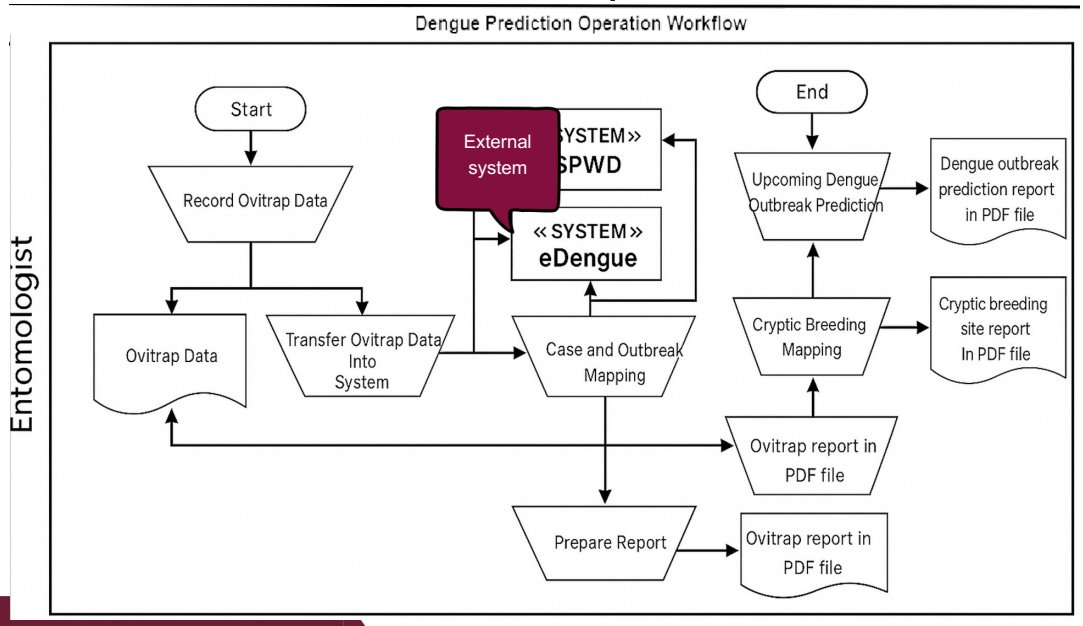
As-Is Swimlane Example



As-Is Swimlane Example



Swimlane Diagram of As-Is Dengue Prediction Operation



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Flowchart Symbols

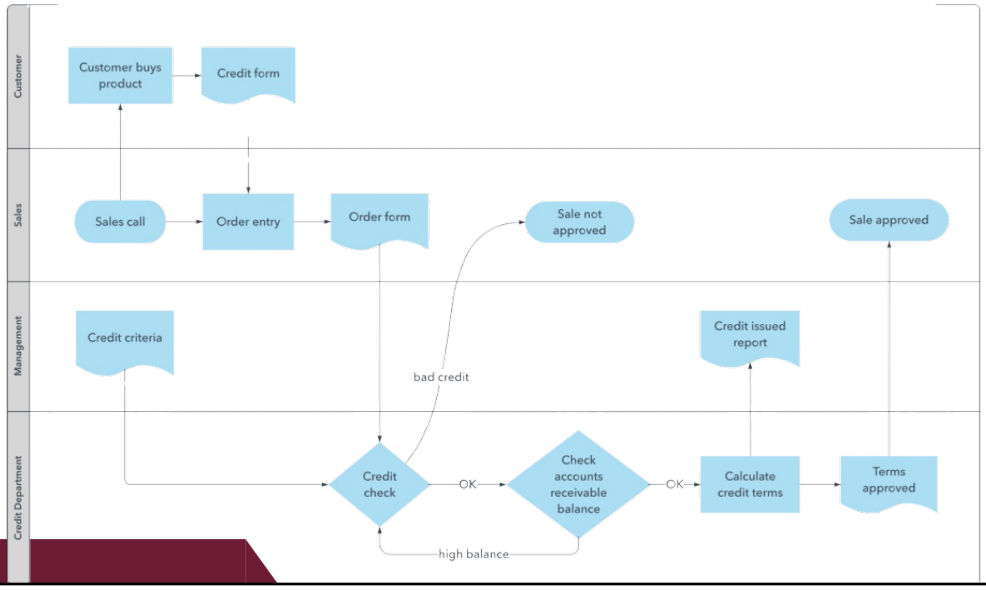
Symbol	Meaning	Symbol	Meaning
	Input/output		Manual operation
	Paper document		Magnetic tape
	Online display		Direct access storage device
	Online input		Flowline
	Punched card		Telecommunications link
	Process		Predefined process
	Terminal		Decision

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Example of Swimlane Diagram

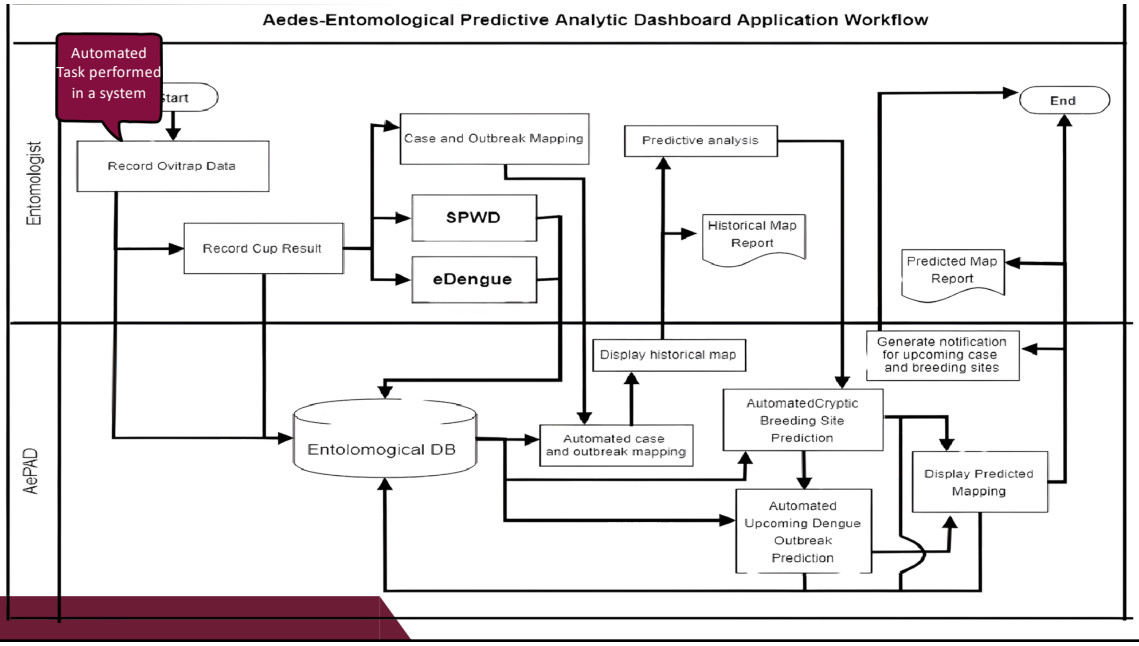
•Read more at:
https://www.lucidchart.com/pages/tutorial/swimlane-diagram



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Swimlane Diagram of Proposed (To-be) Dengue Prediction Operation



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